

B. Tech. End Semester Examination, Session : 2024 - 25
Branch : Computer Science & Engineering / Information Technology
Semester : Fourth
Paper Code : CSUDTT1 / ITUDTT3
DISCRETE MATHEMATICS
SET : A

Time Allowed : 3 Hours

Maximum Marks : 60

- Note :*
- (1) Attempt **SIX** questions in all. **Question Number 1 is Compulsory** and carries **20 Marks**. All other questions carries **08 Marks each**.
 - (2) Only Scientific Calculator is Allowed.

SECTION - A
(Short Answer Type Questions)

1 : Attempt **ALL** parts of question. Each parts carries equal TWO Marks. $10 \times 2 = 20$.

- (A) In a class of 25 students, 12 have taken Mathematics, 8 have taken Mathematics but not Data Science. Find the number of students who have taken Mathematics and Data Science, and those who have taken Data Science but not Mathematics. [CO 1 : BTL 1]
- (B) Let $X = \{a, b, c\}$, Define $f : X \rightarrow X$ such that $f = \{(a, b), (b, c), (c, a)\}$. Find f^4 . [CO 1 : BTL 1]
- (C) What is the number of derangements in a collection of five elements ? [CO 2 : BTL 1]
- (D) How many different ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left neighbor and the same right neighbor ? Explain. [CO 2 : BTL 1]
- (E) Express $p \vee q \vee \neg r$ in terms of NOR ($\downarrow$) operate only. [CO 3 : BTL 1]
- (F) What are the contrapositive, the converse, and the inverse of the conditional statement : "The home team wins whenever it is raining ?" [CO 3 : BTL 1]
- (G) How many transpositions are there of a Permutation Group S_5 ? [CO 4 : BTL 1]
- (H) Prove that : $\overline{\overline{A \cdot \overline{B}} + ABC + A(B + \overline{AB})} = 0$ [CO 4 : BTL 1]
- (I) A graph has 24 vertices and 30 edges. It has five vertices of degree 4, seven pendant vertices, and seven vertices of degree 2. All other vertices have degree 3 or 4. How many vertices of degree 4 are there ? [CO 5 : BTL 1]
- (J) Represent the expression as a binary tree and write the Prefix and Postfix forms of the expression : $A * B - C \uparrow D + E / F$ [CO 5 : BTL 1]

SECTION - B
(Long Answer Type Questions)

Note: Attempt any FIVE questions. Each question carries EIGHT Marks. $5 \times 08 = 40$.

- 2 : (A) Demonstrate the pictorial representation of Partial Order Set. Draw the Diagram and pictorial representation of Partial Order Set of the Divisors of 105. [CO 1 : BTL 3].
- (B) Show that the set $\mathbb{Q}^+$ of all positive rational numbers is Countable. [CO 1 : BTL 2].
- 3 : (A) Consider the five members from a group of twelve to work as a team on a special project. How many distinct five-person teams can be chosen? Suppose two members of the group of twelve insist on working as a pair, any team must contain either both or neither. And suppose two members of the group don't get along and refuse to work together on a team, then how many five-person teams can be formed? Explain it in details. [CO 2 : BTL 4].
- (B) To find g.c.d. of 1701 and 3768. Also express the g.c.d. as a linear combination of 1701 and 3768. [CO 2 ; BTL 5].
- 4 : (A) Show that $(x)(P(x) \vee Q(x)) \implies (x)P(x) \vee (\exists x)Q(x)$ [CO 3 : BTL 4].
- (B) Use the Laws, Show that $[(p \vee q) \implies r] \wedge (\neg p) \implies (q \implies r)$ is a Tautology. [CO 3 : BTL 3].
- 5 : (A) Prove that the Complete Disjunctive Normal Form and Complete Conjunctive Normal Form in three variables in a Boolean Algebra is equal to the unit and zero element of the Boolean Algebra. [CO 4 : BTL 3].
- (B) Show that $(\mathbb{Z}_5, \oplus_5, \otimes_5)$ is a ring where $\oplus_5$ and $\otimes_5$ are the operations of addition and multiplication modulo 5. [CO 4 : BTL 3].
- 6 : Explain the algorithm used to construct Eulerian Path and Cycles. By using that algorithm, find Euler Circuit in the given Graph G (Figure 1) : [CO 5 ; BTL 6].

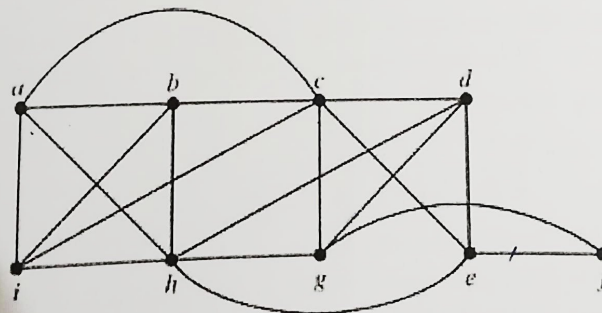


Figure 1: Graph G

aecdef

①

- 7 : (A) Discuss the Chromatic Number ? Also, Find the chromatic number of the graph in Figure 2. [CO 5 : BTL 2].

(M)

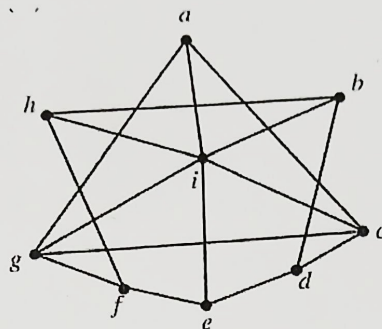


Figure 2: Graph G

- (B) If in a group G , $x^5 = e$, $xyx^{-1} = y^2$ for $x, y \in G$ then show that order of y i.e. $o(y) = 31$. [CO 4 : BTL 5].

8 : Write a short notes on the following :

- (A) The Well Ordering Principle
- (B) The Fundamental Theorem of Arithmetic
- (C) Scope of Quantifiers
- (D) Functionally Complete Set of Connectives

[CO 2 : BTL 1].

[CO 2 : BTL 1].

[CO 3 : BTL 1].

[CO 3 : BTL 1].

Department of Information Technology
School of Studies of Engineering & Technology
Guru Ghasidas Vishwavidyalaya, Bilaspur (C.G.)

B.Tech. IV Semester Examination
Subject: Python for Data Science (Theory)
[ITUDTT1]
SET - II

Time Allowed: 3 hours

Max. Marks 60

Note:-

1. This question paper contains two sections with section – A with 20 and section B with 40 marks respectively.
2. Students are required to attempt questions from both the sections.

I. Fill in the blanks:-

(10 x 2 marks = 20 marks)

- i. _____ function is used to add an element to an empty list.
- ii. A _____ is an unordered collection of elements without duplicates.
- iii. The first argument of the `__init__()` method must be _____.
- iv. Statements written in _____ block are executed irrespective of whether an exception has occurred or not.
- v. After statement `import numpy as np` _____ is the statement to create a 4 x 4 nd array having values ranging 0 to 15 (both inclusive).
- vi. `Sort (matrixname, axis = 0)` is used to sort the elements of numpy array _____ wise.
- vii. The _____ method in pandas is used to access the first 5 rows of a DataFrame.
- viii. A _____ object represents one column of the DataFrame.
- ix. `set_index()` method in pandas is used to _____.
- x. _____ function is used to finally display the graph using `matplotlib.pyplot`.

II. Answer any one question from each of the following:- (8 x 5 marks = 40 marks)

1. Explain dictionary and its use with the help of an example python program.

Or

What are the different types of arguments that a function in python can have?

Explain each with the help of an example program.

2. Explain module loading and execution in python. Write a module with member variables and functions. Illustrative the use of modules with a python program.

Or
Explain diamond problem in python. How it can be solved? Explain with an example program.

3. Write a python program to accept two 3×3 matrices and find their product matrix.

Or
With respect to numpy arrays in python explain the following:-
a. slicing operations in multi-dimensional arrays
b. attributes of an array

4. Write a python program to perform the following:-
i. Create a data frame from python list of tuples (data should contain emp_id, emp_name, salary and date_of_joining fields)
ii. Sort the rows on Date_of_joining column into ascending order.

Or
Create a data frame using csv file (data should contain emp_id, emp_name, emp_age, salary and date_of_joining fields). Also write a query to display all records with emp_age = 20 and salary > 20,000.

5. What is an outlier? How can we identify an outlier and missing data from a data file? Illustrate with example.

Or
Write a program to create a line graph to show the profits of a company in various years. Use appropriate data to create the graph. The graph should also have a title, labels and legends.

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Department of Information Technology
School of Studies (E&T) Guru Ghasidas Vishwavidyalaya Bilaspur, Chhattisgarh

Course: B.Tech. IT

END Semester Examination of B.Tech. IV Semester 2024-25

Year & Semester: 2nd Year, 4th Semester

Subject Code: ITUDTK1
Duration: 3 Hours

Max Marks: 60

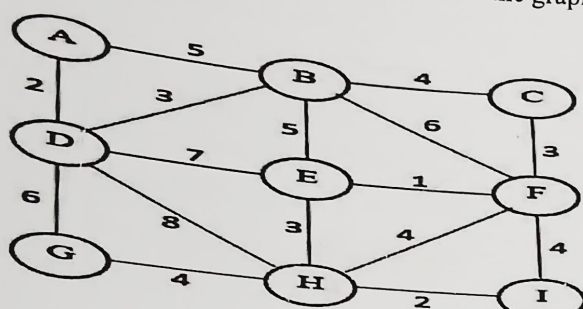
Part-A (Attempt all questions. Each question carries 02 marks) (10x2=20)

Que.1		Marks	CO	BTL
A	What is an Algorithm?	[2]	1	1
B	What are the techniques that are used to solve recurrence relation?	[2]	1	1
C	Explain bi connected components.	[2]	2	1
D	Define the Minimum Spanning Tree.	[2]	2	1
E	Compare Greedy method and Dynamic Programming.	[2]	3	1
F	What is articulation point in graph?	[2]	3	1
G	State Hamiltonian Circuit Problem.	[2]	4	1
H	Define Branch and Bound.	[2]	4	1
I	Define min heap and max heap.	[2]	5	1
J	Differentiate variable length encoding and fixed length encoding	[2]	5	1

Part-B (Attempt any 2 from each question. Each question carries 04 marks) (10x4=40)

		Marks	CO	BTL
Que.2	A. Solve the recurrence relation using Master's Theorem: (i) $T(n) = 3T(n/3) + \sqrt{n}$ (ii) $T(n) = 4T(n/2) + n/\log n$	[4]	1	3
	B. Write merge sort algorithm to sort a set of numbers in ascending order and explain its best, worst and average case time complexity.	[4]	1	3
	C. Write an algorithm for Linear search and discuss its complexity.	[4]	1	3
Que.3	A. Define Greedy knapsack. Find the optimal solution of the Knapsack instance $n = 7$, $M = 20$, $(p_1, p_2, p_3, p_4, p_5, p_6, p_7) = (8, 5, 6, 7, 6, 12, 3)$ and $(w_1, w_2, w_3, w_4, w_5, w_6, w_7) = (2, 10, 8, 7, 6, 4, 11)$.	[4]	2	3
	B. Discuss Draw the portion of state space tree generated by FIFOBB for the following instance of 0/1 knapsack $n = 5$, $M = 12$, $(p_1, p_2, p_3, p_4, p_5) = (10, 15, 6, 8, 4)$ and $(w_1, w_2, w_3, w_4, w_5) = (4, 6, 3, 4, 2)$.	[4]	2	3
	C. Using Huffman Algorithm, construct an optimal binary prefix code for the letters in the given frequency table.	[4]	2	3

Letter	C	E	I	R	S	T	X
Frequency	12	8	18	10	9	5	2

	Draw your Huffman Tree, and Give an optimal binary prefix code for each letter. Give the total number of bits required to encode the string given in the frequency table.																											
Que.4	A. Write down algorithm for Breadth First Search. Explain it with suitable example.	[4]	3	3																								
	B. Compute a minimum cost spanning tree for the graph of figure using prim's algorithm.	[4]	3	3																								
																												
	C. Find Longest Common Subsequence using Dynamic Programming Technique with illustration $X = \{A, B, C, B, D, A, B\}$ and $Y = \{B, D, C, A, B, A\}$.	[4]	3	3																								
Que.5	A. State and explain cook's theorem?	[4]	4	3																								
	B. Explain Satisfiability problem.	[4]	4	3																								
	C. What are P and NP and NP complete problems? Give examples. What is meant by <u>reducibility</u> ?	[4]	4	3																								
Que.6	A. Explain Vertex Cover Problem with help of one example graph.	[4]	5	3																								
	B. Consider the below matrix for assignment problem involving persons and jobs. Explain in detail how branch and bound technique is useful in solving assignment problems.	[4]	5	3																								
	<table border="1" data-bbox="100 1420 1038 1630"><thead><tr><th></th><th>Job1</th><th>Job2</th><th>Job3</th><th>Job4</th></tr></thead><tbody><tr><td>A</td><td>4</td><td>3</td><td>8</td><td>6</td></tr><tr><td>B</td><td>5</td><td>7</td><td>2</td><td>4</td></tr><tr><td>C</td><td>16</td><td>9</td><td>3</td><td>1</td></tr><tr><td>D</td><td>2</td><td>5</td><td>3</td><td>7</td></tr></tbody></table>		Job1	Job2	Job3	Job4	A	4	3	8	6	B	5	7	2	4	C	16	9	3	1	D	2	5	3	7		
	Job1	Job2	Job3	Job4																								
A	4	3	8	6																								
B	5	7	2	4																								
C	16	9	3	1																								
D	2	5	3	7																								
	C. Using Knuth-Morris-Pratt Algorithm match pattern $P = \text{"ababada"}$ in Text $T = \text{"badbabababadaab"}$	[4]	5	3																								

SET-B

B. Tech (Second Year (Fourth Sem) Examination 2024-25
Branch: Common for Chemical Engg. & Information Technology
Sub: Introduction to Information Science, Paper Code: CSUDT01
Time allowed: Three Hours
Maximum Marks: 60

Note: Attempt all sections as directed. Section A is compulsory.

SECTION-A

Note: All questions are compulsory.

I. Multiple Choice Question:

[5x2=10]

1. Entropy of a random variable is [CO-1, BLT-1, Marks-2]
(a) 0 (b) 1
(c) Infinite (d) cannot be determined
2. Which can detect two-bit errors? [CO-2, BLT-1, Marks-2]
(a) Parity Check (b) CRC
(c) Parity & CRC (d) None of these
3. Which of the following is an example of lossless compression? [CO-3, BLT-2, Marks-2]
(a) JPEG (b) BMP
(c) MPEG (d) RLE
4. Which of the following is an example of lossy compression? [CO-4, BLT-2, Marks-2]
(a) JPEG (b) Huffman Coding
(c) BMP (d) Delta Coding
5. Cryptanalysis is used _____? [CO-5, BLT-2, Marks-2]
(a) To find some insecurity in a cryptographic scheme (b) To increase the speed
(c) To encrypt the data (d) To make new ciphers

II. Define the following terms: [5x2=10]

1. Mutual Information [CO-1, BLT-2, Marks-2]
2. Block Codes [CO-2, BLT-2, Marks-2]
3. Lossless Compression [CO-3, BLT-2, Marks-2]
4. Video Image Compression [CO-4, BLT-2, Marks-2]
5. Symmetric Key Cryptography [CO-5, BLT-2, Marks-2]

SET-B

SECTION-B

Note: Attempt any one question from each unit. [8x5=40]

UNIT-I

1. Consider a source $S = \{S_1, S_2, S_3\}$ with $P = \{0.5, 0.25, 0.25\}$. Find

[CO-1, BLT-2, Marks-8]

- (a) Self-information of each message
- (b) Entropy of source "S"

2. Apply Shannon-fano coding for the following message
 $[M] = [M_1, M_2, M_3, M_4]$, $[P] = [0.1, 0.2, 0.3, 0.4]$

[CO-1, BLT-2, Marks-8]

Calculate the value of

- (a) Entropy
- (b) Average Length
- (c) Efficiency
- (d) Redundancy

UNIT-II

3. Define VRC. Explain it with one numerical example.
4. Define the following:-

[CO-2, BLT-2, Marks-8]
[CO-2, BLT-2, Marks-8]

- (a) Parity check
- (b) Single-bit error
- (c) Block codes
- (d) Tree codes

UNIT-III

5. Define Compression. Describe the different types of compression. [CO-3, BLT-2, Marks-8]
6. Write a short note on:- [CO-3, BLT-2, Marks-8]
- (a) CCITT group-3 1D Compression
 - (b) CCITT group-3 2D Compression

UNIT-IV

7. Describe the complete methodology of MPEG compression. [CO-4, BLT-2, Marks-8]
8. Describe any two applications of both video and audio compression. [CO-4, BLT-2, Marks-8]

UNIT-V

9. Describe Diffie-Hellman public key distribution with one numerical example.

[CO-5, BLT-2, Marks-8]

10. Describe the DES algorithm with one numerical example.

[CO-5, BLT-2, Marks-8]

Guru Ghasidas Vishwavidyalaya, SoSE&T
Department of Information Technology
B.Tech (IT 4th Sem) Operating System (ITUDTT2)
Maximum Marks: 60 Time Allowed: 3 Hours

Instructions:

1. Section A is compulsory. Attempt all 10 questions. Each question carries 2 marks. (10 × 2 = 20 marks)
2. In Section B, attempt one question from each unit. Each question carries 8 marks. (5 × 8 = 40 marks)
3. Draw diagrams wherever necessary.
4. Assume suitable data wherever required and justify your assumptions.

Section A (Short Answer Type Questions)

1. What are the main objectives of an operating system?
2. Define file organization. Mention any two file access mechanisms.
3. What is a critical section in process synchronization?
4. Define inter-process communication (IPC) with an example.
5. What is the purpose of a CPU scheduling algorithm?
6. Define turnaround time and waiting time in process scheduling.
7. What is a deadlock? Mention any two conditions necessary for deadlock to occur.
8. What is the purpose of a Resource Allocation Graph (RAG)?
9. Define paging. How does it help in memory management?
10. What is thrashing in virtual memory?

Section B (Long Answer Type Questions)**UNIT I**

- Q1. Explain the objectives and functions of an operating system. How do batch, time-sharing, and real-time operating systems differ?

OR

- Q2. Explain file organization and access mechanisms. What are the services provided by an operating system related to file management?

UNIT II

- Q3. What is a semaphore? How can semaphores be used to solve the producer-consumer problem?

OR

- Q4. Describe the role of the Process Control Block (PCB) in process state transitions. How does the operating system keep track of process states using PCB?

UNIT III

- Q5. Consider the following processes and calculate average waiting time and turnaround time using the SJF (Non-Preemptive) algorithm:

Process	Arrival Time	Burst Time
P1	0	6
P2	1	8
P3	2	7
P4	3	3

OR

- Q6. Compare and contrast preemptive and non-preemptive scheduling algorithms with suitable examples.

UNIT IV

- Q7. Consider the following snapshot of a system using the Banker's Algorithm. The system has five processes (P0 to P4) and three resource types (A, B, C). The following matrices show the Allocation, Maximum, and Available resources:

Allocation Matrix

	A	B	C
P0	0	1	0
P1	2	0	0
P2	3	0	2
P3	2	1	1
P4	0	0	2

Maximum Matrix

	A	B	C
P0	7	5	3
P1	3	2	2
P2	9	0	2
P3	2	2	2
P4	4	3	3

Available Resources: $A = 3, B = 3, C = 2$

Tasks:

- Calculate the Need Matrix.
- Determine whether the system is in a safe state using the Banker's Algorithm.
- If yes, provide a safe sequence. If not, explain why.

OR

- Q8. Draw and explain a Resource Allocation Graph. How can it be used to detect and recover from a deadlock?

UNIT V

- Q9. A process is allocated 4 frames. Its page reference string is:

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2

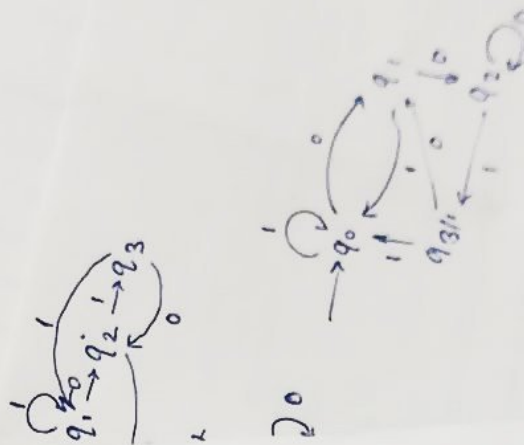
Apply the Least Recently Used (LRU) page replacement algorithm and calculate the number of page faults.

OR

- Q10. What is virtual memory? Explain the concepts of paging and segmentation with suitable diagrams.

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1. Write the difference between read, readline & readlines function.
 2. Explain range function and its use in python.
 3. How are private members in a class created?
 4. Write the two important packages that needs to be imported for drawing a graph from data in python.
 5. What is the use of groupby when writing a SQL statement in python.
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Department of Information Technology, School of Studies of Engineering & Technology,
Guru Ghasidas Vishwavidyalaya (Central University), Bilaspur- CG

Examination date: 8th April 2025

B.Tech 4th Semester IT

Answer any 2 Questions

Subject: Mini Project Viva-Voice Examination
Time: 25 minutes, Max Marks: 15

- Q.1 How does your project stand out from existing solutions or research?
- Q.2 What are the underlying theories or models, and how are they validated in your work?
- Q.3 Can your project be misused? If yes, how would you mitigate that risk?
- Q.4 What is the biggest weakness in your project, and how are you addressing it?
- Q.5 Can you defend why your chosen methodology is better than alternative approaches?
- Q.6 Suppose you had double the time and resources, how would your project design change?